

Student: Toheeb Eji
Date: 12/01/25

Instructor: Aleksandra Gajic
Course: ECON 1BB3: Introductory
Macroeconomics

Assignment: Chapter 3

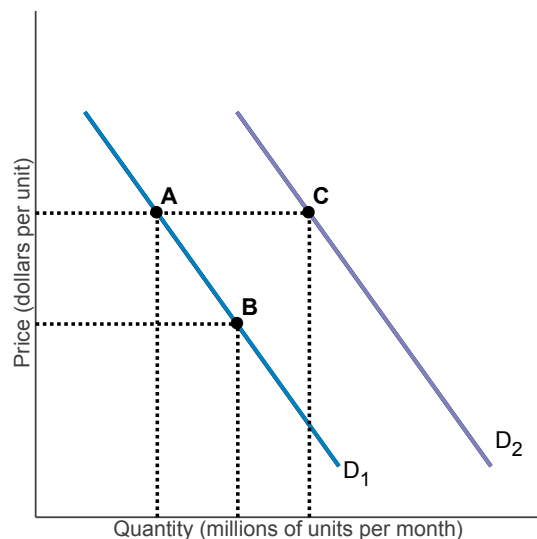
1. Choose the products whose demand is likely to increase rapidly if the following demographic groups increase at a faster rate than the population as a whole: Teenagers.

Select all that apply.

- ☐ A. Ballroom dance classes
- ☒ B. Music downloads
- ☒ C. Text messaging services
- ☐ D. English language courses
- ☐ E. Sesame Street educational software
- ☒ F. Public education
- ☐ G. Prepaid phone cards

2. On the diagram to the right, a movement from A to C represents a

- ☐ A. movement up the demand curve.
- ☐ B. decrease in demand.
- ☐ C. change in quantity demanded.
- ☒ D. change in demand.

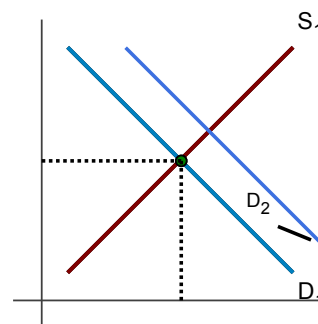


3. The recent recession that began in December 2007 resulted in declining incomes as some people lost their jobs and others were forced to work fewer hours or have their wages reduced. How did this affect the market for fast food restaurants (an inferior good)?

In the figure to the right, show how the recent recession likely affected the market for fast food restaurants.

Using the line drawing tool, draw either a new supply curve (S_2) or a new demand curve (D_2). Properly label your line.

Carefully follow the instructions above, and only draw the required object.



4.

Consider the market for iPhones.

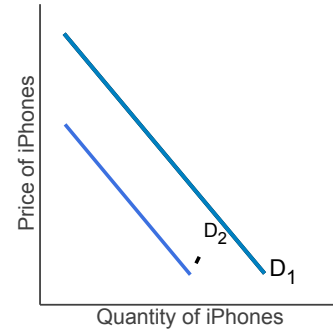
Suppose household income decreases.

Use the line drawing tool to show how this affects the demand for iPhones by drawing a new demand curve. Assume iPhones are a normal good. Properly label this line.

Carefully follow the instructions above, and only draw the required objects.

According to the graph, when income decreases, the demand for iPhones decreases.

Instead, consider an inferior good. If income decreases, then the demand for an inferior good increases.



5.

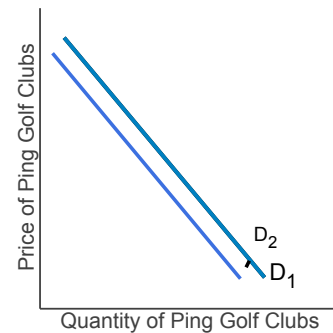
Consider the market for Ping golf clubs.

Suppose the price of memberships at local golf courses increases.

Use the line drawing tool to show how this affects the demand for Ping golf clubs by drawing a new demand curve. Assume memberships at local golf courses and Ping golf clubs are complements. Properly label this line.

Carefully follow the instructions above, and only draw the required objects.

Instead, suppose the price of tennis rackets increases. If tennis rackets and golf clubs are substitutes, then the demand for Ping golf clubs will increase.



6. After World War II in 1945, Canada experienced a "baby boom" as birthrates rose and remained high through the early 1960s. In 2011, the first members of the baby boom generation became older than 65. What effect will this have on the market for medicine?

As the first baby boomers become older than 65, the

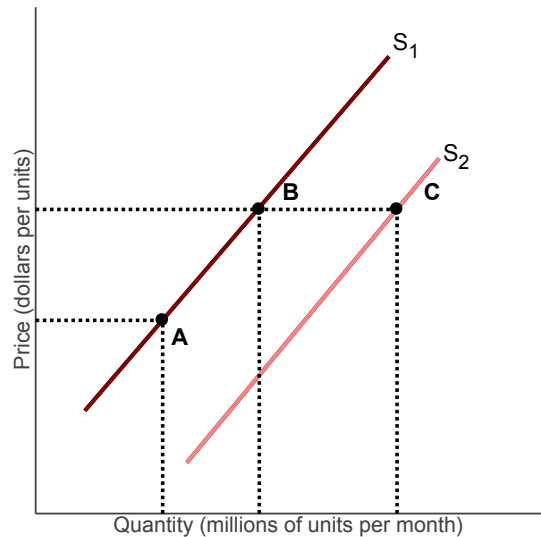
- ☐ A. demand curve for medicine will shift to the left.
- ☐ B. demand curve for medicine will become vertical.
- ☐ C. quantity of medicine demanded will increase.
- ☒ D. demand curve for medicine will shift to the right.
- ☐ E. supply curve for medicine will shift to the right.

7. The difference between a change in supply and a change in the quantity supplied is that the latter is

- ☐ A. determined by the willingness of producers to sell while the former is set by the ability of firms to produce.
- ☐ B. conditional upon a change in the former, but not vice versa.
- ☒ C. produced by a change in the product's own price while the former is caused by a variety of variables other than the product's price.
- ☐ D. shown as a shift in the supply curve while the former is displayed graphically as a movement along a supply curve.

8. On the diagram to the right, a movement from B to C represents a

- ☐ A. decrease in supply.
☐ B. movement down the supply curve.
☒ C. change in supply.
☐ D. change in quantity supplied.



9. Consider the market for the Nissan Xterra.

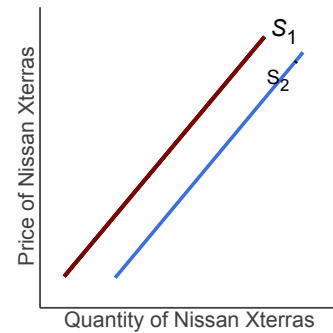
Suppose the price of metal, which is an input in automobile production, decreases.

Use the line drawing tool to show how this affects the supply of Nissan Xteras by drawing a new supply curve. Properly label this line.

Carefully follow the instructions above, and only draw the required objects.

According to the graph, when the price of metal decreases, the quantity of Nissan Xteras supplied at any particular Xterra price increases.

Instead, consider the future price of Xteras. If Nissan believes the future price of Xteras will be higher, then Nissan may reduce supply today.

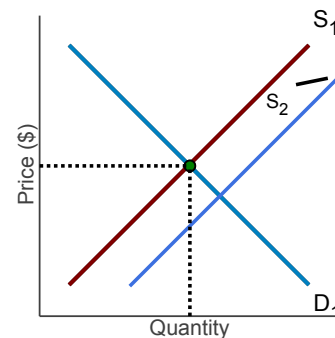


10. Suppose recent changes to immigration laws have increased the supply of immigrant labor, decreasing wages in manual labor occupations. How might this affect the market for construction?

In the figure to the right, show how the wage changes have likely affected the market for construction.

Using the line drawing tool, draw either a new supply curve (S_2) or a new demand curve (D_2). Properly label your line.

Carefully follow the instructions above, and only draw the required object.



11.

Consider the market for Hewlett-Packard (HP) printers, depicted in the figure to the right, where the supply of HP printers has increased from S_1 to S_2 .

What would cause the supply curve for HP printers to shift to the right?

- ☐ A. A decrease in the price of a substitute in production.
- ☐ B. A positive technological change.
- ☐ C. A higher expected future price for HP printers.
- ☒ D. Both a and b.
- ☐ E. All of the above.



12. When economists speak of a surplus, they mean a situation in which

- ☐ A. the market price is above the equilibrium price.
- ☐ B. the quantity supplied exceeds quantity demanded.
- ☐ C. firms have unsold goods piling up.
- ☒ D. all of the above.
- ☐ E. A and B only.

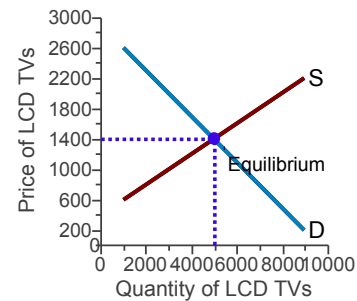
13.

Consider the market for LCD TVs, illustrated in the figure to the right.

Use the point drawing tool to identify the market equilibrium. Properly label this point.

Carefully follow the instructions above, and only draw the required objects.

Suppose instead that the price of LCD TVs is \$2000.00. This will result in a surplus, which will place downward pressure on the price.



14. Refer to the diagram:

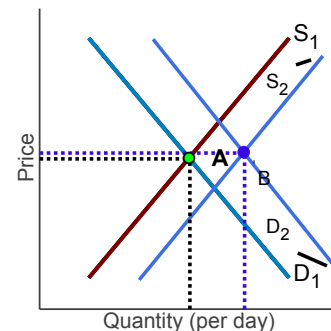
1.) Use the line drawing tool to draw a demand curve that shifts to the right. Label this line ' D_2 '.

2.) Use the line drawing tool to draw a supply curve that shifts to the right by less than the demand line. Label this line ' S_2 '.

3.) Use the point drawing tool to identify the new point of equilibrium. Label this point

Carefully follow the instructions above, and only draw the required objects.

With these shifts, the equilibrium price will increase.



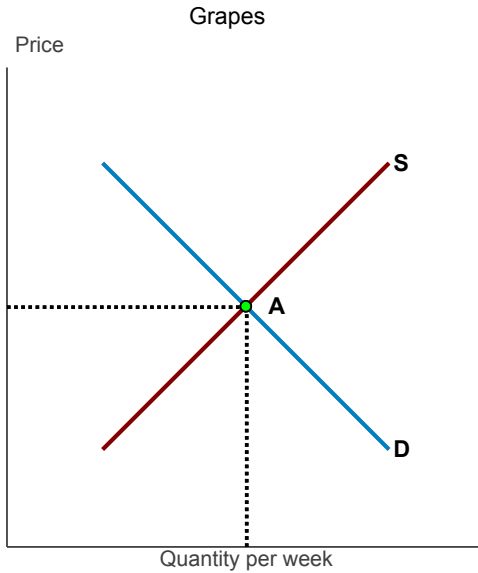
15. The demand for grapes is highest during summer and lowest during winter. Yet grape prices are normally lower in summer than in winter. What must be happening to the supply of grapes, from winter to summer, for the equilibrium price to fall?

¹ Click on the icon to experiment with inward and outward shifts in supply and demand. Note the change in equilibrium price and quantity.

- ☐ A. The supply does not change.
- ☒ B. The supply increases more than the demand increases.
- ☐ C. The supply for grapes decreases.
- ☐ D. The supply increases, but less than the demand increases.

1: Graph/Chart

Click anywhere in the following diagram and use the sliders to simulate shifts in either or both lines.



16.

An article in the *Wall Street Journal* notes that during early 2009 the demand for Internet advertising was declining at the same time that the number of Internet sites accepting advertising was increasing. After reading the article, a student argues:

"From this information, we know that the price of Internet ads should fall, but we don't know whether the total quantity of Internet ads will increase or decrease."

Is the student's analysis correct? Illustrate your answer with a demand and supply graph.

Source: Martin Peers, "Future Shock for Internet Ads?" *Wall Street Journal*, February 17, 2009.

Assume the equilibrium price of Internet advertisements is initially P_1 and the equilibrium quantity is Q_1 .

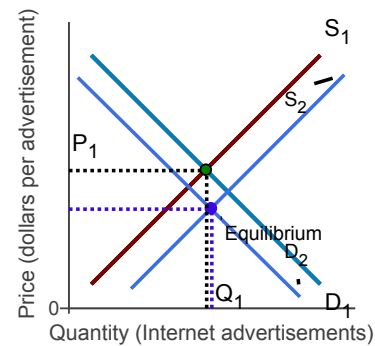
1.) Use the line drawing tool to draw a new supply curve for Internet advertisements (S_2) and a new demand curve for advertisements (D_2). Properly label the lines.

2.) Use the point drawing tool to indicate the market equilibrium price and quantity. Label this point 'Equilibrium'.

Carefully follow the instructions above, and only draw the required objects.

The student's analysis is

- ☐ A. correct because the price of Internet ads will decrease and the quantity of Internet ads will decrease if the shift in supply is larger and will increase if the shift in demand is larger.
- ☒ B. correct because the price of Internet ads will decrease and the quantity of Internet ads will decrease if the shift in demand is larger and will increase if the shift in supply is larger.
- ☐ C. incorrect because the quantity of Internet ads will not change.
- ☐ D. incorrect because the quantity of Internet ads will unambiguously decrease when demand decreases.
- ☐ E. incorrect because the quantity of Internet ads will unambiguously increase when demand decreases.



17. A student writes the following: "Decreased production leads to a higher price, which in turn decreases demand." Is the student's reasoning accurate? Briefly explain.

- ☐ A. The student's reasoning is incorrect. A higher price increases the demand.
- ☐ B. The student's reasoning is correct. Decreased production shifts the supply curve to the left, which increases the price and shifts the demand curve to the left.
- ☒ C. The student's reasoning is incorrect. The quantity demanded decreases, but the demand curve does not shift.
- ☐ D. The student's reasoning is incorrect. Decreased production leads to a lower price.

18.

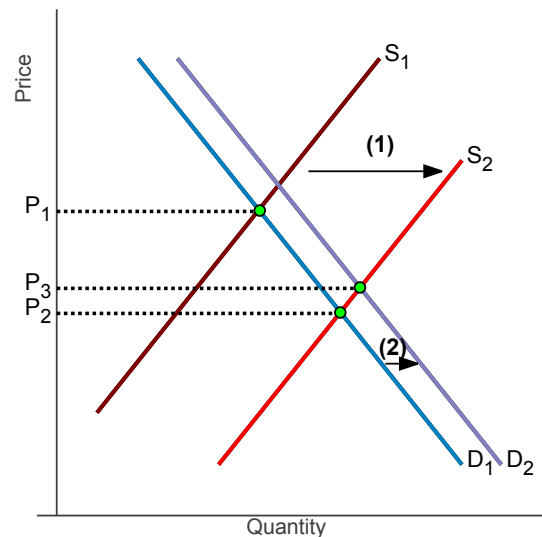
A student was asked to draw a demand and supply graph to illustrate the effect on the tablet computers market of a fall in the price of displays used in tablet computers, holding everything else constant

She drew the graph to the right and explained it as follows:

"Displays are an input to tablet computers, so a fall in the price of displays will cause the supply curve for tablets to shift to the right (from S_1 to S_2). Because this shift in the supply curve results in a lower price (P_2), consumers will want to buy more tablets, and the demand curve will shift to the right (from D_1 to D_2). We know that more tablets will be sold, but we can't be sure whether the price of tablets will rise or fall. That depends on whether the supply curve or the demand curve has shifted farther to the right. I assume that the effect on supply is greater than the effect on demand, so I show the final equilibrium price (P_3) as being lower than the initial equilibrium price (P_1).

Is the student's argument correct?

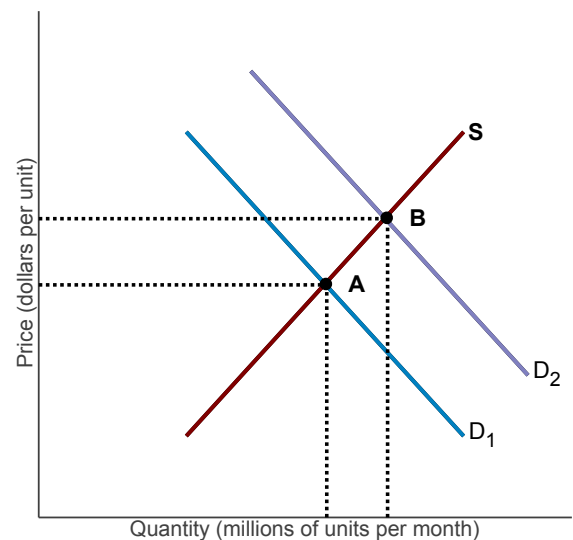
- ☐ A. Yes, her explanation is pretty good.
- ☒ B. No, because a shift of demand curve will not happen.
- ☐ C. No, because there will be not a shift of the supply curve, but a rightward movement along it.
- ☐ D. No, because there will be not a shift of the demand curve, but a leftward movement along it.



19.

In the diagram to the right, when demand increases, a shortage develops at the original price.

Equilibrium price will rise and equilibrium quantity will rise as a new equilibrium is established.



20. Consider the market for college textbooks illustrated in the figure.

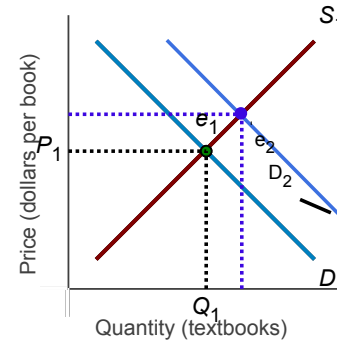
Suppose a recent economic recession has resulted in increased tastes for learning.

In the figure, show how this has likely affected the market for college textbooks.

1.) Using the line drawing tool, draw either a new supply curve (S_2) or a new demand curve (D_2). Properly label your line.

2.) Using the point drawing tool, indicate the new market equilibrium (e_2).

Carefully follow the instructions above, and only draw the required objects.



21. Suppose the price of a complement to LCD televisions rises. What effect will this have on the market equilibrium for LCD TVs?

The equilibrium price of LCD TVs will

- ☐ A. decrease and the equilibrium quantity will increase.
- ☐ B. not change and the equilibrium quantity will not change.
- ☒ C. decrease and the equilibrium quantity will decrease.
- ☐ D. increase and the equilibrium quantity will decrease.
- ☐ E. increase and the equilibrium quantity will increase.

22. Consider the market for MP3 players, illustrated in the figure to the right. The market is initially in equilibrium at a price of \$70 and at a quantity of 250 (thousand) players.

Suppose there is a positive change in technology.

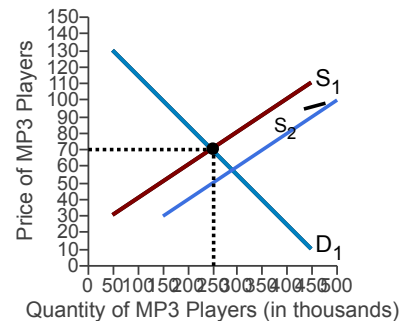
Use the line drawing tool to show how this affects the MP3 market by adding either a new supply curve or a new demand curve.

Carefully follow the instructions above, and only draw the required objects.

As a result of the positive change in technology, the new equilibrium price will be lower, and the new equilibrium quantity will be higher.

Suppose instead that the demand curve shifts to the right. Then, relative to the initial market equilibrium,

- ☐ A. the new equilibrium price would be lower and the new equilibrium quantity would be lower.
- ☒ B. the new equilibrium price would be higher and the new equilibrium quantity would be higher.
- ☐ C. the new equilibrium price would be lower and the new equilibrium quantity would be higher.
- ☐ D. the new equilibrium price would be higher and the new equilibrium quantity would be lower.



23. Suppose the equilibrium price and equilibrium quantity of gold both decrease. Which of the following would produce such a change?

- ☒ A. The market demand curve for gold could have decreased.
- ☐ B. The market supply curve for gold could have decreased or the market demand curve for gold could have increased.
- ☐ C. The market supply curve for gold could have increased.
- ☐ D. The market demand curve for gold could have increased.
- ☐ E. The market supply curve for gold could have decreased or the market demand curve for gold could have decreased.

24. In a linear demand equation, what economic information is conveyed by the intercept on the price axis?

In a linear demand equation, the intercept on the price axis tells us

- ☐ A. the lowest price at which there is no shortage.
- ☐ B. the price at which the quantity demanded will be the highest.
- ☐ C. the equilibrium price.
- ☐ D. the quantity demanded when the price is zero.
- ☒ E. the lowest price at which the quantity demanded will be zero.

Similarly, the intercept on the price axis in a linear supply equation gives the

- ☐ A. the lowest price at which there is no surplus.
- ☐ B. the equilibrium price.
- ☐ C. the price at which the quantity supplied will be the highest.
- ☒ D. the highest price at which the quantity supplied will be zero.

25.

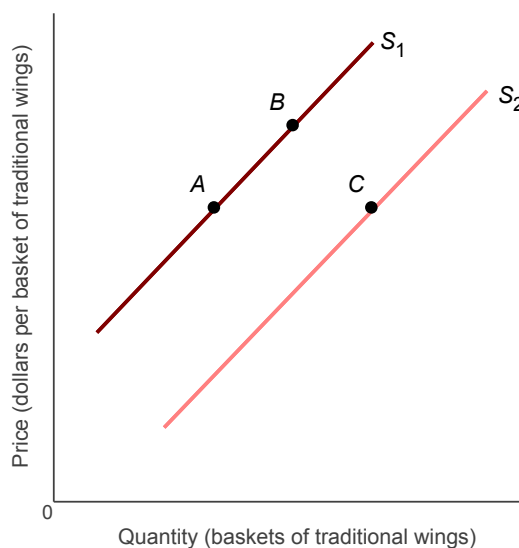
Suppose that the curves in the figure to the right represent two supply curves for traditional wings (basket of six) at Buffalo Wild Wings.

The movement from point A to B on S_1 is caused by

- ☒ A. an increase in the price of baskets of traditional wings.
- ☐ B. a decrease in the price of baskets of traditional wings.
- ☐ C. a decrease in input prices.

Indicate which of the following would cause a movement from point A to C. (Check all that apply.)

- ☒ A. An increase in worker productivity.
- ☐ B. An increase in the number of buyers.
- ☒ C. A decrease in input prices.
- ☐ D. An increase in the product's price.



26. Identify whether each of the following statements is true or false.

a. If the demand and supply for a product both increase, the equilibrium quantity of the product must also increase.

☒ A. True

☐ B. False

b. If the demand and supply for a product both increase, the equilibrium price of the product must also increase.

☐ A. True

☒ B. False

c. If the demand for a product decreases and the supply of the product increases, the equilibrium price of the product may increase or decrease, depending on whether supply or demand has shifted more.

☐ A. True

☒ B. False

27.

Imagine that the table shows the quantity demanded² of Allbirds shoes at five different prices in 2021 and in 2022.

Which of the following variables could cause the demand for Allbirds shoes to change as indicated from 2021 to 2022? (Check all that apply.)

☒ A. A decrease in the price of a substitute good

☐ B. The expectation that Allbirds shoes will rise in price

☒ C. A decrease in the number of buyers

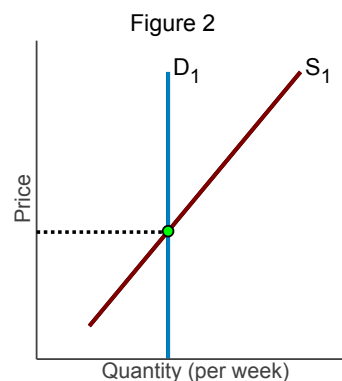
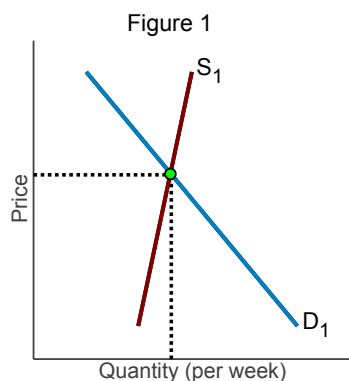
☐ D. An increase in the price of Allbirds shoes

Price	Quantity Demanded 2021	Quantity Demanded 2022
\$160	8,000	7,000
170	7,500	6,500
180	7,000	6,000
190	6,500	5,500
200	6,000	5,000

2: Definition

Quantity demanded is the amount of a good or service that a consumer is willing and able to purchase at a given price.

28. Below are the supply and demand functions for two markets. One of the markets is for BMW automobiles, and the other is for a cancer-fighting drug, without which lung cancer patients will die.



Which diagram represents the cancer-fighting drug?

Figure 2

Why?

☐ A. Market demand reflects the availability of many substitutes.

☒ B. Market demand reflects the availability of few substitutes.

☐ C. Market supply reflects the availability of few substitutes.

